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New antimony fluorooxoborates with strong birefringence and unprecedented structural characterisation†

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Fluorooxoborates constitute a rich source of optical crystals due to their structural diversity and excellent performance. Antimony fluorooxoborates with stereochemically active lone pairs of electrons still have not been found, although the first antimony borate was discovered several years ago. In this study, we have achieved the successful synthesis of the first antimony(III) fluorooxoborate with an unprecedented $[B_2O_4F]_\infty$ chain, namely SbB_2O_4F. Remarkably, SbB_2O_4F shows strong birefringence (0.171@1064 nm) and short UV cutoff edges (about 220 nm) according to calculations. The birefringence of SbB_2O_4F mainly originates from the highly distorted $[SbO_4]$ groups.

New ultraviolet (UV) optical crystals, especially for birefringent crystals, play important roles in modern laser and polarization technology owing to their vital function in modulating the polarization of light.[1–5] Borates constitute one of the best material systems for investigating UV optical crystals because of their diversity in structural chemistry and excellent performance. When the fluorine atom is introduced into the coordination environment of the boron atom in borates, fluorooxoborates are formed. The introduction of the F atom breaks the symmetry of the $[BO_4]$ tetrahedron in the structure and the anisotropy is enhanced. Thus, the polarizability anisotropy of $[BO_xF_{4-x}]$ (x = 1, 2, 3) groups is larger than that of $[BO_4]$ groups. Systematic studies of crystal structures of a series of borates and fluorooxoborates have shown them displaying excellent optical properties.[6–24] Some fluorooxoborates, including $NH_4B_4O_6F$, NaB_4O_6F, CsB_4O_6F, BiB_2O_4F, $SnB_2O_3F_2$, and $SrB_5O_7F_3$, have been presented in the literature and reported to show strong optical responses[25–32] Moreover, combining oxyfluoride units with borate groups has been reported to lead to abundant anionic groups of borates in the form of isolated clusters, chains, layers, or frameworks.[33–40] Using a classical dipole–dipole interaction model, the average polarizability anisotropy of chain-like $[B_nO_{2n+1}]$ ($n \geq 1$) groups polymerized by $[BO_3]$ anionic groups, such as $[B_2O_5]$, $[B_3O_7]$, and the $[BO_2]_\infty$ chain, have been shown to be greater than that of isolated $[BO_3]$. In addition, the dangling bond is eliminated in these cases, which causes the UV cutoff edge of compounds to be blue-shifted in favor of a larger bandgap. Hence, $[BO_2]_\infty$ chains are attracting a lot of interest in efforts at designing crystals with attractive optical performances. The current 1D chains in the borate system can be divided into six types: those composed of only $[BO_3]$ units, only $[BO_4]$ units, only $[BO_2F_2]$ units, a combination of $[BO_3]$ and $[BO_4]$ units, a combination of $[BO_3]$ and $[BO_3F]$ units, and a combination of $[BO_2F_2]$, $[BO_4]$ and $[BO_3F]$ units. The new 1D chains are still worth investigating in the borate system.[6]

The elements frequently used as chemical constituents in the borate system are alkali metals and alkaline earth metals. The work on contributions to birefringence from various chemical constituents is mainly focused on the anionic [B–O] framework, whereas there are very few reports of birefringence data on metal cations with lone pairs of electrons, especially for Sb^{3+}. From the perspective of cation selection, the stereochemical activity of lone pair electrons of antimony atoms leads to favorable properties for optical applications. The materials composed of a metal with a lone pair of electrons show large birefringence.[41–47] According to our survey, only a few Sb(V)-containing borate compounds have been reported, including $LnSb_2BO_8$ (Ln = Sm, Eu, Gd, or Tb), $K_3Sb_4BO_{13}$, $ASbOB_2O_5$ (A = K or Rb), and $A_2SbB_3O_8$ (A = Na, K, or Rb), and $BaSb_2B_4O_{12}$,[48–51] and only two Sb(III)-containing borate compounds have been reported, namely, SbB_3O_6 and SbB_5O_9, which show strong birefringence;[52,53] the scarcity here may be due to Sb^{3+} being easily oxidized to Sb^{5+}. A large birefringence enhancement (of about 60 times) has been realized due to the

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substitution of alkali metal by antimony in isostructural $K_3SbP_2O_9$ and α-$NaSb_3P_2O_{10}$ as reported by our group.[46]

In this study, through reactions in a closed system to avoid oxidization, we successfully synthesized the first antimony(III) fluorooxoborate, namely SbB_2O_4F. Notably, the 1D $[B_2O_4F]_\infty$ chain has been observed as the fundamental building block (FBB) in fluorooxoborates for the first time. In addition, compared to other fluorooxoborates, SbB_2O_4F exhibits strong birefringence (Δn = 0.171@1064 nm) and is a potential UV birefringent material.

SbB_2O_4F crystallizes in the centrosymmetric space group $P\bar{1}$, which is a low-symmetry triclinic space group. Relevant crystallographic data and bond valence sum (BVS) for each atom in the asymmetric unit are shown in Tables S1–S4 in the ESI.† As shown in Fig. S1 (ESI†), an inductively coupled plasma element analysis proves the presence of Sb, B, O, and F, consistent with the results of single-crystal XRD analysis. The energy-dispersive X-ray spectroscopy and BVS results are consistent with the X-ray-determined structure. As shown in Fig. 1a and b, it exhibits a 3D framework composed of $[BO_3]$ triangles, $[BO_3F]$ tetrahedra, and $[SbO_4]$ polyhedra. B atoms are located in two independent crystallographic positions, which are coordinated with three and four atoms, respectively. As shown in Fig. 1c, $[BO_3]$ triangles and $[BO_3F]$ tetrahedra are interconnected by corner sharing to form indefinitely long $[B_2O_4F]_\infty$ chains, which run along the *a*-axis with isolated $[SbO_4]$ tetrahedra serving as bridges to form a 3D framework (Fig. 1d).

To conduct an in-depth investigation of the structure–property relationship, we performed first-principles calculations based on density functional theory methods. As shown in Fig. S2 (ESI†), the indirect band gap derived from these calculations is 4.38 eV for SbB_2O_4F. Simultaneously, we performed calculations using the HSE06 method, with these results showing that SbB_2O_4F has a large band gap of 5.60 eV, which corresponds to a cutoff edge of about 220 nm. To simulate accurately the interaction between electrons in the crystal field, we used scissors to calibrate the GGA calculation result to the experimental gap or HSE06 calculation results.

To further analyze the origin of the strong birefringence of SbB_2O_4F, total density of states, partial density of states (PDOS), and band structure for SbB_2O_4F were investigated and are shown in Fig. S2 (ESI†). It is common that at about the Fermi level, electronic transitions between states play a major role in the optical performances of compounds. According to these calculations, the top region of valence bands (VBs) is mainly occupied by the 5s and 5p states of Sb, the 2p states of O, and the 2p states of F. Further, the bottom region of conduction bands (CBs) is mainly occupied by the 5p states of Sb accompanied by B 2p orbitals. According to PDOS, $[SbO_4]$ polyhedra and π-conjugated $[BO_3]$ units make a major contribution to the optical properties of SbB_2O_4F. The introduction of fluorine atoms in alkali/alkaline earth metal fluorooxoborates results in a lack of terminal oxygen atoms and produces a large band gap because the terminal fluorine atom is more electronegative than oxygen.[27] Similarly in SbB_2O_4F, the introduction of fluorine has beneficial effects on the band gap, although the band gap is mainly affected by Sb. We performed first-principles calculations to characterize the birefringence of SbB_2O_4F, with the results showing a high birefringence value of 0.171 @1064 nm (Fig. S3, ESI†).

To date, more than 3000 cases of borates but fewer than 100 types of inorganic fluorooxoborates have been reported. To the best of our knowledge, SbB_2O_4F is the first antimony(III) fluorooxoborate to have been reported. Most anionic B–O/F frames are 0D isolated clusters of borates. Borates with 0D isolated B–O/F units with a low polymerization degree always accompany isolated $[BO_2]$, $[BO_3]$, $[BO_4]$, $[BOF_3]$, and $[BF_4]$ units.

As shown in Fig. 2, all available indefinitely long 1D B–O/F chains in borates can be classified into six types.[6] In this study, the 1D $[B_2O_4F]_\infty$ chain (consisting of BO_3 and BO_3F) is observed for the first time in borates, which greatly enriches the structural chemistry of borates. According to the composition of the 1D chain, which is formed only by the $[BO_3]$, $[BO_4]$, and $[BO_2F_2]$ groups by sharing oxygen atoms, they can be classified as the $[BO_2]_\infty$ chain, the $[BO_3]_\infty$ chain, and the $[BOF_2]_\infty$ chain.[54–60] Metal cations combine with these 1D chain units to form several excellent materials, such as $LiBO_2$ and $Ca(BO_2)_2$ composed of the $[BO_2]_\infty$ chain.

In SbB_2O_4F, Sb^{3+} cations bond to neighboring O^{2-} anions, forming $[SbO_4]$ groups with a pyramidal shape, with the Sb 5s lone pair oriented toward the free space, indicating a strong anisotropicity of $[SbO_4]$. The microscopic second-order susceptibility tensors from groups are canceled out as a result of the formed antiparallel arrangement, which implies that they crystallize in a centrosymmetric space group. In the electron localization function

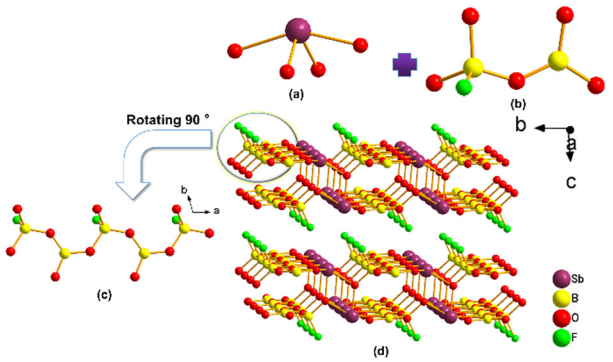

Fig. 1 (a)–(c) Local structures of the (a) coordination environment of the Sb atom, (b) coordination environment of B and (c) 1D chain for $[B_2O_4F]_\infty$. (d) Overall structure of SbB_2O_4F.

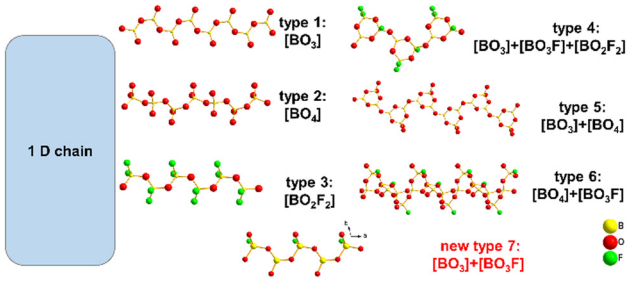

Fig. 2 Structures of borates with 0D and 1D B–O/F anionic frameworks.

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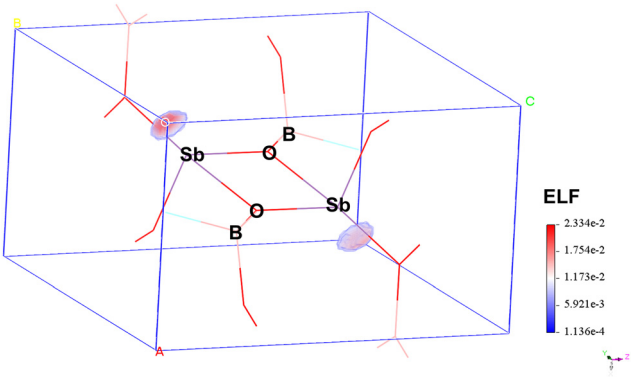

Fig. 3 Electron localization function diagrams of [Sb–O] polyhedra in SbB_2O_4F.

diagram produced (Fig. 3), we see the presence of lone-pair electrons of Sb^{3+}. As shown in Fig. S2b (ESI†), (Sb 5s–O 2p)* and Sb 5p arise at about the Fermi level, resulting in a sterically active asymmetric density on Sb, which also proves the presence of lone pairs. $[SbO_4]$ polyhedra with the stereochemically active lone pair have large distortions, which benefit the optical anisotropy, causing a large contribution to birefringence. For the anion group, π-conjugated $[BO_3]$ primitives show much better polarizability anisotropy than non-π-conjugated $[BO_2F_2]$ primitives. Thus, the π-conjugated $[BO_3]$ group also synergistically contributes to the birefringence.

The birefringence levels of compounds containing six other chains is shown in Table S5 (ESI†). Due to the low anisotropy, tetrahedral groups such as $[2BO_4 + BO_3F]$ units in BiB_2O_4F and $[BO_2F_2]$ units in Cc-$BaBOF_3$ show little structural distortion, and hence contribute little to the birefringence. The crystals with the $[BO_2]_\infty$ chain usually exhibit strong birefringence, due to the high anisotropy of the $[BO_3]$ unit. By introducing Sb^{3+} with a stereochemically active lone pair into the tetrahedral primitive, a substantial increase in birefringence results.

By combining the $[B_2O_4F]_\infty$ chain with the system based on antimony, SbB_2O_4F compound is obtained. A comparison of this compound with the reported birefringent crystals of inorganic oxygenates based on antimony is shown in Fig. 4.[60,61] SbB_2O_4F shows strong birefringence in the reported birefringent crystals of inorganic oxygenates based on antimony.

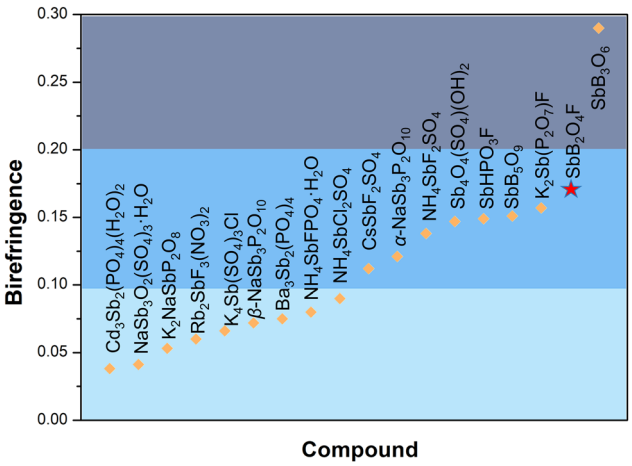

Fig. 4 Comparison of the reported birefringence of crystals of inorganic oxygenates based on antimony.

In summary, a novel antimony(III) fluorooxoborate, namely SbB_2O_4F, was obtained. The 1D $[B_2O_4F]_\infty$ chain is first observed as FBBs in fluorooxoborates for the first time. Remarkably, SbB_2O_4F displays strong birefringence (0.171@1064 nm) and a short UV cutoff edge (about 220 nm). This study has enriched the structural chemistry of fluorooxoborates and provided a rarely investigated material, which may cause the finding of new materials displaying strong birefringence.

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Conflicts of interest

The authors declare no conflicts of interest.

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